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/*****
*   COMPANY                      : TAKEDA
*   PROTOCOL/STUDY/PROJECT      : ShortCourse
*   AUTHOR                      : OS Adjusted Analysis Working Group
*   DATE                        : 08Sep2025
*   PROGRAM NAME                : case_study_2way_IPCW_clean.sas
*   LOCATION                    : /.../.../...
*   VERSION/ PLATFORM           : SAS V9.4 / LINUX
*   PURPOSE                     : Program to Perform Inverse Probability of Censoring Weighting (IPCW)
*                               : Analysis - 2 way.
*   MACROS CALLED               :
*   INPUT                      :
*   OUTPUT                     :
*   PARAMETERS (Optional)      :
*                               :
*   NOTES (Optional)           : ASA Traveling Course, September to October 2025
*
*****/

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options noquotelenmax;
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** define relevant directories ;
libname ads '/bdm/tbos/AP24534/studies/201/csr/adhoc/ShortCourse/Clean_SAS_Programs';
%include "/bdm/tbos/AP24534/studies/201/csr/adhoc/ShortCourse/Demo/macro_adj_pvalue.sas";
%include "/bdm/tbos/AP24534/studies/201/csr/adhoc/ShortCourse/Demo/macro_km_plot.sas";

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data final;
    set ads.final_casestudy1;
    by usubjid analdy;
    month2=sqrt(month);
    hgb2=sqrt(hgb);
    psa2=sqrt(psa);
    *** For IPCW, both altrx and censor into account;
    censor2=0;
    if (censor=1 or altrx=1) then censor2=1;
    if last.usubjid then analdy=survtime;

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run;
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ods exclude all;
*****/
*** creating weight for IPCW;

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*****;

/* Model 1, weight for non-informative censoring, numerator */
proc logistic data=final;
  where analdy<=althdy or althdy=.;
  class trt region pdbase pandrog racecat alkcat ldhcat visceral acthfl antinefl
    stra_factor hgbcats;
  model censor2=trt region pandrog racecat ecog month month2 alkcat ldhcat
    visceral acthfl antinefl;
  output out=model1 p=altv_0;
run;

/* Model 2, weigh for non-informative censoring, denominator */
proc logistic data=final;
  where analdy<=althdy or althdy=.;
  class trt region pdbase pandrog racecat alkcat ldhcat visceral acthfl antinefl
    stra_factor hgbcats;
  model censor2=trt region pandrog racecat ecog pd highpsa month ecogt psa hgb
    hgb2 psa2 month2 alkcat ldhcat visceral acthfl antinefl highpsa;
  output out=model2 p=altv_w;
run;

**** for IPCW analysis, may consider incorporating censoring weight;
proc sort data=model1;
  by usubjid analdy die;
run;

proc sort data=model2;
  by usubjid analdy die;
run;

data main_w2;
  merge model1 model2;
  by usubjid analdy die;

  /* variables ending with _0 refer to the numerator of the weights
  Variables ending with _w refer to the denominator of the weights */
  *** compute stablized weight;

  /* reset the variables for a new patient */
  if first.usubjid then do;
    k1_0=1;

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        k1_w=1;
    end;
    retain k1_0 k1_w;

    /* Inverse probability of censoring weights, they are IPTW in the MSM model */
    if analdy<althdy or althdy=. then do;
        k1_0=k1_0*altv_0;
        k1_w=k1_w*altv_w;
    end;

    /* patients that start alternative therapy in the current month */
    else if analdy=>althdy then do;
        k1_0=0;
        k1_w=k1_w;
    end;

    /* Stabilized weights only */
    stabw=k1_0/k1_w;
run;

*** model check;

proc sort data=main_w2 out=main_w2a;
    by trt;
run;

proc univariate data=main_w2a noprint;
    var stabw;
    by trt;
    where stabw >0;
    output out=chk_w2a_trt n=n mean=mean std=std min=min max=max p99=p99 p1=p1 sum=sum;
run;

proc univariate data=main_w2a noprint;
    var stabw;
    where stabw >0;
    output out=chk_w2a_all n=n mean=mean std=std min=min max=max p99=p99 p1=p1 sum=sum;
run;

ods exclude none;

proc print data=chk_w2a_all;

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        title "check weight: mean should be near 1, IPCW weight";
        var n mean std min max;
run;

**** max is not too big, No need to rescale stabw for huge ones;

data main_w12;
    set main_w2;
    by usubjid analdy;
    mid=1;
run;

proc sort;
    by trt mid;

data chk_w2a_trt2;
    set chk_w2a_trt;
    by trt;
    mid=1;
    keep trt p99 mid;
run;

data main_w12aa;
    merge main_w12 chk_w2a_trt2;
    by trt mid;
    q=floor(month/3);

    if stabw>p99 then
        stabw=min(p99*(1+ranuni(analdy)), stabw);
run;

data wtplot;
    set main_w12aa;

    if stabw ne 0 then
        logwt=log(stabw);
    q=floor(month/3);
    keep usubjid visit q trt logwt stabw;
run;

proc sort data=wtplot;
    by q trt;

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run;

ODS GRAPHICS ON;
proc boxplot data=wtplot;
    title 'After rescaling, check weight: mean should be near 1, IPCW weight';
    plot logwt*q;
run;
ods graphics off;

proc univariate data=main_w12aa noprint;
    var stabw;
    where stabw >0;
    *** as they are censored obs;
    output out=chk_w2aa n=n mean=mean std=std min=min max=max p99=p99 p1=p1 sum=sum;
run;

proc print data=chk_w2aa;
    title 'After rescaling, check weight: mean should be near 1, IPCW weight';
    var n mean std min max;
run;

proc sort data=main_w12aa out=main_w2;
    by usubjid analdy;
run;

*** reorganize data for proc phreg;

data main_w2b;
    set main_w2;
    by usubjid analdy;
    *** data error;
    a=lag(analdy);

    if first.usubjid then do;
        start=0;
        stop=analdy;
    end;
    else do;
        start=a;
        stop=analdy;
    end;

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        if stop=start then stop=stop+0.001;

        if stabw=. then stabw=1;
run;

*** final IPCW model;
ods select ModelANOVA ParameterEstimates;
ods output ParameterEstimates=stat_ipcw;
proc phreg data=main_w2b covs(aggregate);
    class trt(order=freq) region pdbase;
    model (start, stop)*die(0)=trt /rl;
    id usubjid;
    weight stabw;
    strata region pdbase;
    title "IPCW Cox model for OS";
    title2 "Using proc phreg, counting process data format for multiple records";
run;

*****;
*** Call macro to calculate z-score and p-value from weighted Log-rank test
*****;
%macro_adj_pvalue(inds=main_w2b          /* Final weighted analysis data */
                  ,method=IPCW          /* Weighted method */
                  );

*****;
*** Call macro to plot adjusted KM curves
*****;
%macro_km_plot(inmain=main_w2b          /* Final weighted analysis data */
               ,instratds=stat_ipcw     /* Final statistic data contains stratified p-value, HR and 95% CI */
               ,inunstratds=stat_uspv_ipcw /* Final statistic data contains unstratified p-value */
               ,method=IPCW             /* Weighted method */
               );

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